



Retail Insights: Customer, Product & Campaign Analysis



Executive Summary

This report aims to highlight business performance from three main perspectives: overall sales performance, customer and product behavior, and the impact of the trial campaign on store performance. The business recorded total sales of \$1.93 million through approximately 265,000 transactions and 72,000 unique customers, with an average transaction value of \$7.30. Overall, the analysis shows a relatively balanced and stable sales pattern, reflecting a solid demand base.

In terms of customer behavior analysis, the Mainstream segment accounted for the largest share of sales, with approximately \$750,000 in sales, while the Older Singles/Couples life-stage segment recorded the highest sales, with total sales of \$402,000.

From a product behavior perspective, the Kettle brand led with revenue of \$390,000, reflecting the brand's significant contribution to the business's total revenue.

From another perspective, the most significant difference was observed in the results of the trial campaign. The comparison with the control stores showed three clearly different outcomes: Store 77 experienced a 72.8% decline in sales, while Store 86 recorded a 4.5% increase in sales, and Store 88 achieved a significant increase of 217.0%. This growth was not limited to sales, but was also accompanied by a 161.32% increase in the number of customers and a 17.90% increase in the average number of transactions per customer.

Overall, the results indicate that the effectiveness of the campaign was related to the characteristics of each individual store. Therefore, before expanding the campaign, the characteristics of the stores that achieved strong results should be identified, while the reasons for weak performance in other stores should be investigated before generalizing the strategy.



Business Context

This report aims to focus on understanding what drives revenue and where and how the impact of business initiatives is reflected. To achieve this, transaction data were analyzed, in addition to customer and product characteristics and store characteristics, which facilitates the interpretation of financial results through purchasing behavior and customer base composition.

The analysis also takes an additional step in evaluating the impact of the trial campaign on three stores. Instead of relying solely on the change in sales of the trial store, we compared it with a similar control store, providing a better basis for evaluating changes in sales, number of customers, and average number of transactions per customer during the trial period.

Business Objectives

Accordingly, the analysis aims to:

- Evaluate overall sales, customer, and transaction performance.
- Identify the customer segments and life stages that contribute the most to revenue.
- Identify the brands and products with the greatest impact on sales.
- Measure the impact of the trial campaign on store performance.
- Compare the results of the trial stores with the control stores.
- Determine the extent to which the effectiveness of the campaign varies across stores.
- Translate the findings into recommendations that can be used to support future business decisions.



Data Review

Regarding the structure of the data on which this analysis was based, it consisted of two main data sources: purchase transaction data and customer attribute data.

The transaction data included information such as store, customer, date, product, quantity, and sales value, while the customer data provided information about the characteristics of the customers themselves.

LYLTY_CARD_NBR was used as the key to merge the two sources, providing a database that enabled performance analysis from both financial and behavioral perspectives.

Before starting the analysis phase, I first carried out the data cleaning stage by examining data quality through the removal of duplicates and missing values and reviewing descriptive statistics. An outlier was also identified in product quantity and excluded from the analysis to avoid its illogical impact on the results.

To transform the raw data into comparable indicators, the date was prepared for monthly analysis, classifications were created for brands and pack sizes, and the data was aggregated at the store-month level to evaluate key metrics such as sales, customers, transactions, and average transactions per customer.

Overall Performance

The overall performance of the business shows that it remained relatively stable throughout the analysis period, with 264,836 transactions recorded from 72,637 unique customers and total sales of approximately \$1.93 million. The average transaction value was \$7.30, while the average quantity was 1.91 units, indicating that the business relies on a high volume of purchases with relatively limited individual quantity and value.

When analyzing the source of revenue, the Mainstream segment ranked first, generating the highest sales at \$750.7 thousand (38.8%), followed by Budget at \$676.2 thousand (35.0%), while Premium contributed \$506.2 thousand (26.2%). This means that nearly three-quarters of revenue comes from Mainstream and Budget, making them the primary drivers of overall business performance and revenue.

Sales also show a relative concentration by life stage, with Older Singles/Couples leading with \$402.4 thousand in sales, followed by Retirees at \$366.5 thousand and Older Families at \$352.5 thousand. At the product level, Kettle was the highest-contributing brand to revenue, generating \$390.2 thousand, while Dorito Corn Chp Supreme 380g ranked first among individual products, with sales of \$39.1 thousand. Therefore, the overall performance picture indicates the presence of a large customer base, but with a clear and limited concentration in the sources of revenue.

Understanding these concentrations is important before moving to the evaluation of the campaign, as they provide the baseline through which any subsequent changes in store performance can be interpreted.



Customer & Segment Analysis

The analysis concluded that revenue does not come from all customers at the same level. Mainstream customers accounted for the largest share of sales at \$750.7 thousand, followed by Budget at \$676.2 thousand, while Premium ranked third at \$506.2 thousand. Therefore, approximately three-quarters of revenue comes from Mainstream and Budget.

When analyzing life stages, Older Singles/Couples led sales with \$402.4 thousand, followed by Retirees and Older Families.

Even with these differences between the segments, the average transaction value remained very similar, indicating that the size of activity within each segment is the most important factor behind the differences in their contribution.

Therefore, the key commercial opportunity is not only to target higher-value customers, but also to target segments with larger customer bases in strategic planning, particularly Mainstream and Budget, as well as the highest-contributing age groups.



Product & Brand Analysis

When looking at sales from a product perspective, a limited group of brands emerges as a key driver of revenue. Kettle leads the top with sales of \$390.2 thousand, while Doritos and Smiths generated \$240.6 thousand and \$224.7 thousand, respectively. Overall, the three brands represent approximately 44% of total sales.

However, the picture becomes more complex and the interpretation more insightful when moving to sales at the individual product level. Dorito Corn Chp Supreme 380g achieved the highest sales among individual products, with \$39.1 thousand, even though Doritos is not the highest-performing brand overall. This confirms that it is not accurate to assess performance based solely on brand-level classification.

The results highlight the importance of maintaining high-contributing products and brands while monitoring individual product performance separately to identify where the true growth and profit opportunities are concentrated and can be leveraged.



Trial Campaign Evaluation

Before examining the campaign results, it was necessary to establish a method to determine whether the store's performance after the campaign had actually changed, rather than simply reflecting natural differences between stores or occurring by chance. Therefore, the pre-campaign period was reviewed and sales patterns between stores were compared. Through this comparison, three stores were identified as suitable control benchmarks: 71 for Store 77, 155 for Store 86, and 134 for Store 88.

Instead of comparing all stores with one another in a general manner, which does not guarantee the validity of the results, this approach provided the precise basis on which the campaign results analysis for the relevant months was built.



Campaign Monthly Performance

When examining the results during the defined campaign period, three different outcomes emerge. For Store 77, sales were lower than those of the control store, meaning that the campaign did not appear to provide a clear performance advantage. For Store 86, the picture remained relatively stable, with the difference between the two stores remaining limited. Store 88, however, followed a completely different trajectory from the previous two cases, significantly outperforming its control store and ending with an overall difference of 4,757.55.

This makes Store 88 the clearest case for investigating whether specific characteristics of the store, customers, or products contributed to this outcome.



The Impact of the Campaign on the Number of Customers

The evaluation of the campaign was not limited to sales, but also included the number of customers as a complementary indicator. Store 77 recorded 196 customers compared with 238 in its control store, Store 71, representing a difference of -42 customers or a relative decrease of 17.65%. Store 86 recorded 249 customers compared with 233 in Store 155, representing a positive difference of 16 customers or an increase of 6.87%. In contrast, Store 88 recorded 317 customers compared with 194 in Store 134, representing a difference of 123 customers and an increase of 63.40%.

These results show that differences in sales were, in some cases, accompanied by clear differences in customer base size, particularly in Store 88, while the decline in customers at Store 77 coincided with a decline in sales.



Statistics

To determine whether the observed differences in sales reflected more than random fluctuations, Welch's independent samples t-test was conducted between the results of the trial stores and their control stores during the campaign period. Store 77 showed a T-value of -20.108 and a P-value of 0.00001 , indicating a statistically significant difference. Store 88 also showed a T-value of 18.190 and a P-value of less than 0.00001 , indicating a strong statistically significant difference. In contrast, Store 86 recorded a T-value of 0.991 and a P-value of 0.35671 , which is above the 0.05 significance level. Therefore, there is insufficient statistical evidence to establish a significant difference in sales between Store 86 and its control store.



Overall Campaign Findings

Based on the three main indicators, it is clear that the trial campaign produced varying results across stores, and the experiment cannot be reduced to a single conclusion applicable to all locations. Store 88 recorded the strongest performance, achieving a 217.03% increase in sales compared with its control store, accompanied by a 63.40% increase in the number of customers and a 17.90% increase in average transactions per customer. In contrast, Store 77 recorded a 72.77% decline in sales, along with a 17.65% decrease in the number of customers and a 19.13% decrease in average transactions per customer. Store 86 showed more moderate results, with sales increasing by 4.52% and the number of customers increasing by 6.87%, while average transactions per customer decreased by 3.34%.

Therefore, the results indicate that campaign response is highly dependent on the characteristics and context of each store.



Recommendations

Recommendation 1: Selective Expansion

Based on the significant variation in results, it is recommended not to apply the campaign uniformly across all stores. Instead, a selective expansion approach should be adopted, starting with stores that exhibit characteristics similar to Store 88. Expansion should be preceded by a more detailed analysis of the factors associated with its performance to determine whether the results can be replicated across other locations.

Recommendation 2: Analyze the Drivers of Store 88's Success

It is recommended to conduct an additional diagnostic analysis of Store 88, focusing on products, customers, and purchasing patterns during the campaign period, with the aim of identifying the factors that contributed to the increase in sales alongside the growth in the number of customers and average transactions per customer.

Recommendation 3: Diagnose Store 77

It is recommended to conduct an analysis of Store 77 to identify the potential reasons behind the decline in sales, number of customers, and average transactions compared with its control store. This could include examining the most influential products, customer patterns, and sales distribution.



Limitations

The data provide a strong picture of the differences between stores, but they do not reveal everything that occurred within the stores during the five-month period. We know that Store 88 outperformed, Store 77 declined, and Store 86 remained close to its control store, but the data alone cannot explain all the reasons behind these outcomes. Therefore, the project identifies a strong signal that warrants further investigation, but it does not claim that the campaign was the sole factor responsible for every observed change.



Conclusion

This project presents an integrated model for applying data analytics to performance evaluation and decision-making support, starting with data preparation and integration using Python, followed by the creation of pivot and summary tables for the results in Excel, and ultimately the development of an interactive Dashboard using Power BI.

This process enables the transition from raw data to interpretable metrics, contributing to a more comprehensive evaluation of the trial campaign by comparing three trial stores with control stores during the period from February to June 2019.

The results showed clear differences between the stores; Store 88 achieved the strongest performance compared with its control store, while Store 77 recorded a clear decline, and Store 86 demonstrated a more moderate performance.

Thus, the project demonstrates how routine operational data can be transformed into strategic insights that help identify opportunities, solve problems, and maximize benefits.